HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and implemented **Java Spring Boot** REST APIs to support new feature development, focusing on clear contract definitions and efficient data exchange.
- Developed and maintained data extraction and transformation scripts using **Python** and **pandas** to populate analytical data stores for reporting needs.
- Executed small-scale technology processes within a laboratory setting, applying principles of **bioprocessing** and **separation sciences** to optimize manufacturing workflows.
- Collaborated with analytical scientists to execute process development studies, contributing to the characterization and improvement of biotherapeutic manufacturing processes.
- Applied knowledge of **bioprocess engineering** and **biochemistry** to contribute to process development projects, ensuring alignment with project objectives.
- Participated in team discussions and experimental design, proactively contributing to a collaborative environment focused on scientific innovation and process optimization.
- Containerized laboratory applications using **Docker**, streamlining deployment and ensuring consistent execution across development and testing environments.

SKILLS

Languages – Java, Python, SQL, R

Frameworks/Libraries – Spring Boot, pandas, NumPy, scikit-learn

Cloud/Infrastructure – AWS (EC2, S3, Lambda, CloudWatch), Docker

Databases – PostgreSQL, MySQL, Oracle

Tools/Platforms – Jenkins, Git, Jira, Confluence, Splunk, Dynatrace, Power BI, Tableau, RStudio

Domain Knowledge – Bioprocessing, Separation Sciences, Biochemistry, Process Development, Manufacturing Optimization, Data Analytics, BI Dashboards

PROJECTS

Bioprocess Data Analysis and Optimization

- Developed Python scripts to extract and analyze data from laboratory instruments, identifying key parameters impacting process efficiency.
- Created interactive **Power BI** dashboards to visualize experimental results and track process performance metrics for scientific review.
- Applied statistical methods from MSc coursework to analyze experimental outcomes, generating insights for process refinement.

Java-based Laboratory Automation Service

- Engineered a **Java Spring Boot** service to automate data logging and sample tracking within a research laboratory environment.
- Implemented RESTful APIs for interaction with laboratory equipment and data repositories, ensuring seamless data flow.
- Containerized the service using **Docker** for simplified deployment and management on lab servers.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)